

COVID-19 Analysis Project Documentation

1. Project Overview

The COVID-19 Data Analysis project focuses on analyzing the impact and spread of COVID-19 using historical COVID-19 data. The project involves cleaning and transforming raw data using Microsoft Excel and creating interactive dashboards and visualizations using Microsoft Power BI.

The analysis provides insights into confirmed cases, deaths, recoveries, active cases, and the progression of COVID-19 across different countries and time periods.

2. Project Objective

- Analyze the spread of COVID-19 over time.
- Identify countries/regions with high numbers of cases.
- Analyze death cases.
- Understand COVID-19 trends using time-series analysis.
- Calculate important KPIs and percentages.
- Create an interactive dashboard for easier analysis.
- Present complex COVID-19 data in a simple visual format.

3. Dataset

Information	Details
Dataset	covid-19_rawdata
Data Source	Source link (click here)
Format	Excel / CSV
Number of Records	60 cols & 999+ rows
Key Fields	Continent, location, date, total_cases, new_cases, total_deaths, median_age.

4. Tools & Technologies

- **Microsoft Excel** — Data cleaning and preprocessing
- **Microsoft Power BI** — Data visualization and dashboard creation
- **Power Query** — Data transformation
- **DAX** — Calculated measures and KPIs

5. Data Cleaning & Preprocessing

Data Cleaning Process

1. Imported the raw COVID-19 dataset into Excel.
2. Selected required columns. Dropped some unnecessary columns.
3. Filtered the date to latest date only i.e. 7/20/2021.
4. Removed null values from the continent, total_cases, new_cases, total_deaths, median_age.
5. Changes naming of columns for standardization i.e. continent to Continent, location to Country, date to Date, total_cases to Total Cases, new_cases to New Cases, total_deaths to Total Deaths, median_age to Median Age.
6. Changes name of the sheet to Covid Final.
7. Creating instances of the same dataset. Selected required columns. Dropped some unnecessary columns.

6. Power BI Dashboard

Dashboard Components

KPI Cards

- Total Cases
- Total Deaths
- New Cases

Charts

- **Area:** Total Cases by Country. Top 10 filtering in Country by Total Cases.
- **Donut/Pie Chart:** Total death by Continent.
- **Donut/Pie Chart:** New Cases by Continent.
- **Area:** Total Cases by Continent.

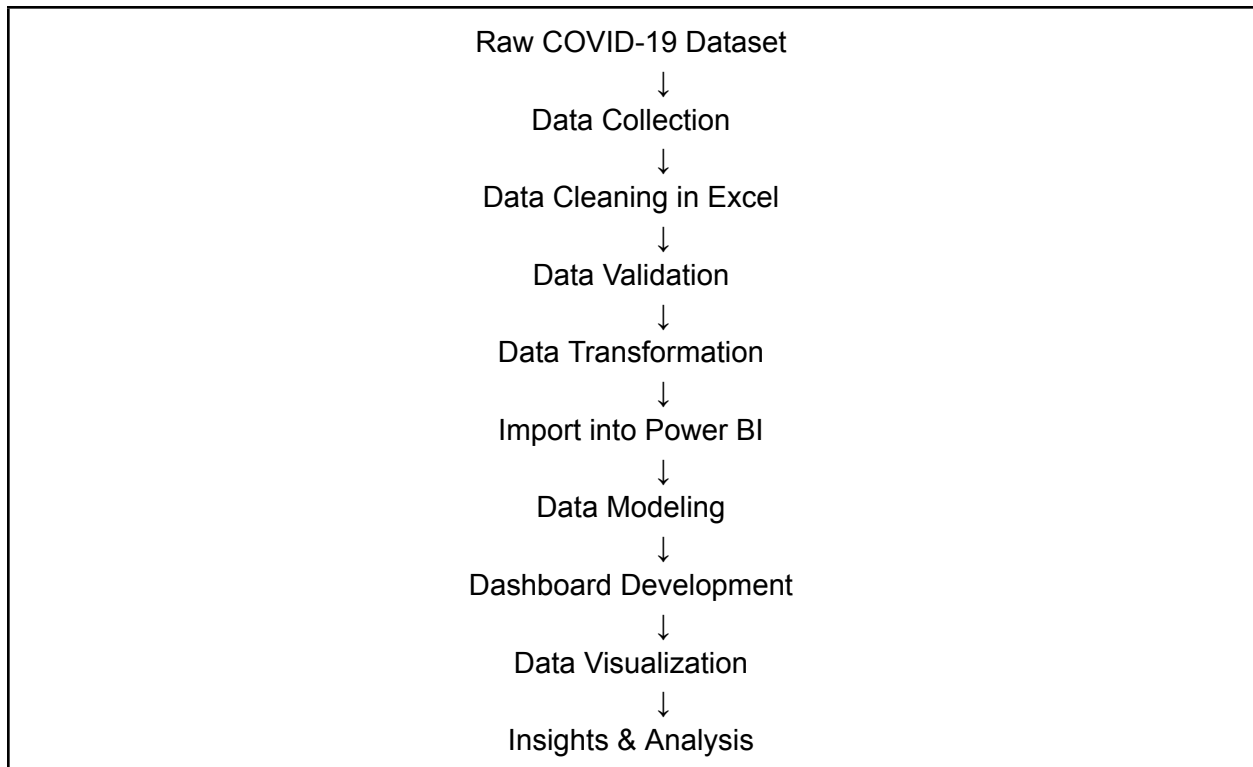
- **Line Chart:** COVID-19 cases over time. Median Age by Continent and Country. Filter to Country selected India, United States, United Kingdom, Brazil, France, Russia, Turkey, Argentina, Columbia, Italy.
- **Line Chart:** COVID-19 cases over time. Total Cases by Year and Country.

7. Dashboard Insights

Key Insights

- The number of confirmed cases changed significantly throughout the analyzed period.
- COVID-19 cases were concentrated differently across countries and regions.
- Some countries recorded substantially higher confirmed cases than others.
- The relationship between Total Cases, New Cases and Total Death changed over time.
- Time-based analysis helped identify major increases and decreases in reported cases.

8. Project Workflow



9. Key Findings

1. Identified the overall progression of COVID-19 cases over the selected period.
2. Compared COVID-19 cases across countries and regions.
3. Analyzed the relationship between Total Cases, New Cases and Total Death.
4. Identified periods with significant changes in reported cases.
5. Developed an interactive Power BI dashboard to make the analysis easier to understand.

10. Business / Analytical Value

The dashboard provides a centralized view of COVID-19 statistics and enables users to explore trends across different years and geographical regions. Interactive filtering and visualizations make it easier to identify patterns, compare regions, and understand changes in COVID-19 cases over time.

11. Project Challenges

- Handling missing or inconsistent data.
- Removing unnecessary records.
- Standardizing country/region names.
- Selecting appropriate visualizations.
- Creating meaningful KPIs.
- Presenting large amounts of data in a simple dashboard.

12. Conclusion

The COVID-19 Data Analysis project demonstrates the use of Microsoft Excel for data cleaning and Microsoft Power BI for data visualization and interactive dashboard development. The project helped transform raw COVID-19 data into meaningful insights through data preprocessing, analysis, KPI development, and visualization.

13. Future Improvements

As this is an academic learning project, no further updates or future enhancements are currently planned.

14. Project Skills Demonstrated

For your portfolio/resume, you can finish with:

Skills Demonstrated:

Microsoft Excel · Data Cleaning · Data Preprocessing · Data Analysis · Power BI · Data Visualization · Dashboard Development · KPI Analysis · Business Intelligence

If you used them, you can additionally add:

Power Query · DAX · Data Modeling